

=== running linear_supervised.py ===

Qwen/Qwen2-VL-7B-Instruct + Contrast Pairs + Supervised Methods (revised)

```
#####  
# CATEGORY: OBJECT_DETECTION  
#####
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EXTRACTING HIDDEN STATES: OBJECT_DETECTION

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✓ Found cached hidden states!

Loading from: hidden_states_cache_final/cache_object_detection_1306_qwen2_ccs_aligned.npz

Loaded: pos=(1306, 3584), neg=(1306, 3584)

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SUPERVISED LOGISTIC REGRESSION

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Dataset split:

Train: 783 samples (402 pos, 381 neg)

Test: 523 samples (251 pos, 272 neg)

Hidden dim: 3584

Logistic regression accuracy: 82.8%

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SUPERVISED LINEAR PROBE

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Dataset split:

Train: 783 samples (402 pos, 381 neg)

Test: 523 samples (251 pos, 272 neg)

Hidden dim: 3584

Results:

Train Accuracy: 90.5%

Test Accuracy: 86.6%

Positive samples: 87.3% (251 samples)

Negative samples: 86.0% (272 samples)

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COMPARISON SUMMARY

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Logistic Regression: 82.8%

Supervised Linear Probe: 86.6%

✓ COMPLETE: object_detection

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#####  
# CATEGORY: ATTRIBUTE_RECOGNITION  
#####
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EXTRACTING HIDDEN STATES: ATTRIBUTE_RECOGNITION

✓ Found cached hidden states!

Loading from:

hidden_states_cache_final/cache_attribute_recognition_3366_qwen2_ccs_aligned.npz

Loaded: pos=(3366, 3584), neg=(3366, 3584)

SUPERVISED LOGISTIC REGRESSION

Dataset split:

Train: 2019 samples (1030 pos, 989 neg)

Test: 1347 samples (653 pos, 694 neg)

Hidden dim: 3584

/gpfs/home6/mdemirev/VisionCCS/vision_ccs/venv/lib/python3.11/site-packages/sklearn/

linear_model/_logistic.py:599: ConvergenceWarning: lbfgs failed to converge after 100 iteration(s)
(status=1):

STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).

You might also want to scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

n_iter_i = _check_optimize_result(

Logistic regression accuracy: 77.9%

SUPERVISED LINEAR PROBE

Dataset split:

Train: 2019 samples (1030 pos, 989 neg)

Test: 1347 samples (653 pos, 694 neg)

Hidden dim: 3584

Results:

Train Accuracy: 89.3%

Test Accuracy: 82.0%

Positive samples: 83.9% (653 samples)

Negative samples: 80.1% (694 samples)

COMPARISON SUMMARY

Logistic Regression: 77.9%

Supervised Linear Probe: 82.0%

✓ COMPLETE: attribute_recognition

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#####  
# CATEGORY: SPATIAL_RECOGNITION  
#####
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EXTRACTING HIDDEN STATES: SPATIAL_RECOGNITION

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✓ Found cached hidden states!

Loading from: hidden_states_cache_final/cache_spatial_recognition_976_qwen2_ccs_aligned.npz
Loaded: pos=(976, 3584), neg=(976, 3584)

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SUPERVISED LOGISTIC REGRESSION

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Dataset split:

Train: 585 samples (289 pos, 296 neg)
Test: 391 samples (199 pos, 192 neg)
Hidden dim: 3584

Logistic regression accuracy: 78.5%

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SUPERVISED LINEAR PROBE

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Dataset split:

Train: 585 samples (289 pos, 296 neg)
Test: 391 samples (199 pos, 192 neg)
Hidden dim: 3584

Results:

Train Accuracy: 87.7%
Test Accuracy: 76.2%
Positive samples: 76.9% (199 samples)
Negative samples: 75.5% (192 samples)

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COMPARISON SUMMARY

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Logistic Regression: 78.5%
Supervised Linear Probe: 76.2%

✓ COMPLETE: spatial_recognition

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FINAL RESULTS SUMMARY

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Category	LogReg	Linear Probe
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object_detection	82.8%	86.6%
attribute_recognition	77.9%	82.0%
spatial_recognition	78.5%	76.2%
Average	79.7%	81.6%

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